Accuracy of MapBiomas Collection 8 land use and land cover classification in the São Luís intermediate region, Brazil

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This report is built from `3_report_en.ipynb`, which re-runs the analysis notebook `2_acuracia_sa23z.ipynb` on the data in the repository github.com/LambdaGeo/acuracia-slz. Every number, table and figure is computed when the report is built. It summarizes the peer-reviewed article, published in Portuguese: doi:10.33360/geonordeste.v36i.21680. Please cite the article.

Abstract. MapBiomas publishes annual land use and land cover maps for Brazil together with national accuracy statistics, but these statistics do not cover several classes that matter on the coast of Maranhão, such as Mangrove, Apicum (hypersaline tidal flat) and Beach and Dune. We assessed the accuracy of MapBiomas Collection 8 for 2022 at legend level 2 in chart group SA-23-Z, which covers about 72% of the São Luís intermediate region. Of the 712 MapBiomas validation samples in the chart group, 655 samples in 11 classes remained after removing classes with too few samples. Based on sample counts, overall accuracy was 71.6%. The overall error of 28.4% splits into 7.6% quantity disagreement and 20.8% allocation disagreement, against 9.4% allocation disagreement nationally. Cohen's Kappa was 0.62, which corresponds to substantial agreement on the Landis and Koch scale. Beach and Dune reached 100% user's and producer's accuracy. Mangrove had a high producer's accuracy (93%) but a 56% commission error: all 18 of its commission errors were Forest Formation samples mapped as Mangrove. Grassland was the least accurate class. Apicum could not be assessed because it had only three reference samples.

Keywords: MapBiomas; accuracy assessment; land use and land cover; remote sensing; Maranhão.

1. Introduction

Land use and land cover maps underpin environmental monitoring, territorial planning and conservation. In Brazil, the MapBiomas project produces annual maps from Landsat imagery, classified with random forest in Google Earth Engine, for every year since 1985 (Souza et al., 2020). MapBiomas also releases its validation samples and accuracy statistics. The statistics are reported for the country and for each biome, and they leave out several level-2 classes, including Mangrove, Apicum, Beach and Dune, Mining and Urban Area.

These classes are central to the São Luís intermediate region, in the north of the state of Maranhão. The region has 73 municipalities and about 70,800 km², and its coast holds some of the largest mangrove and dune areas in Brazil, including the Lençóis Maranhenses. This study assesses the accuracy of MapBiomas level-2 classes for this region, so that local users know how far the maps can be trusted there.

2. Data and methods

2.1 Reference samples

MapBiomas assesses accuracy with about 75,000 spatially independent samples distributed over Brazil and stratified by IBGE chart groups and slope classes. Each sample was labelled by three interpreters, with majority voting and a fourth reviewer for ties (Parente et al., 2021). Sample sizes per chart group were set for a maximum margin of error of 5% at a 95% confidence level. To keep that design intact, the analysis uses one whole chart group rather than clipping samples to the region boundary: SA-23-Z, which covers about 72% of the São Luís intermediate region and contains 712 samples.

The MapBiomas Google Earth Engine code was adapted only to restrict the export to SA-23-Z (`1_export_gee_input.js`). It returns, for each sample, the reference label and the Collection 8 classification for 2022.

2.2 Pre-processing

Class codes were converted and aggregated to legend level 2 with the MapBiomas reconciliation rules. Classes that could not be assessed reliably were then removed: classes with fewer than 10 reference samples (Silviculture, Apicum and Other Non-Vegetated Areas), Floodable Forest, which had no reference samples, and Herbaceous Sandbank Vegetation (Restinga Herbácea), which had reference samples but was never mapped. As in the MapBiomas code, only classes present in both the classification and the reference were kept. This left **655 samples in 11 classes**.

2.3 Accuracy metrics

From the confusion matrix we computed overall accuracy; user's accuracy and its complement, the commission error; and producer's accuracy and its complement, the omission error. Following MapBiomas, the overall error was split into quantity disagreement and allocation disagreement (Pontius and Millones, 2011). We also report Cohen's Kappa, interpreted with the scale of Landis and Koch (1977), despite its known limitations, for comparison with other studies. All metrics in Sections 3.1 to 3.3 are computed from sample counts, without area weighting.

3. Results

3.1 Per-class accuracy

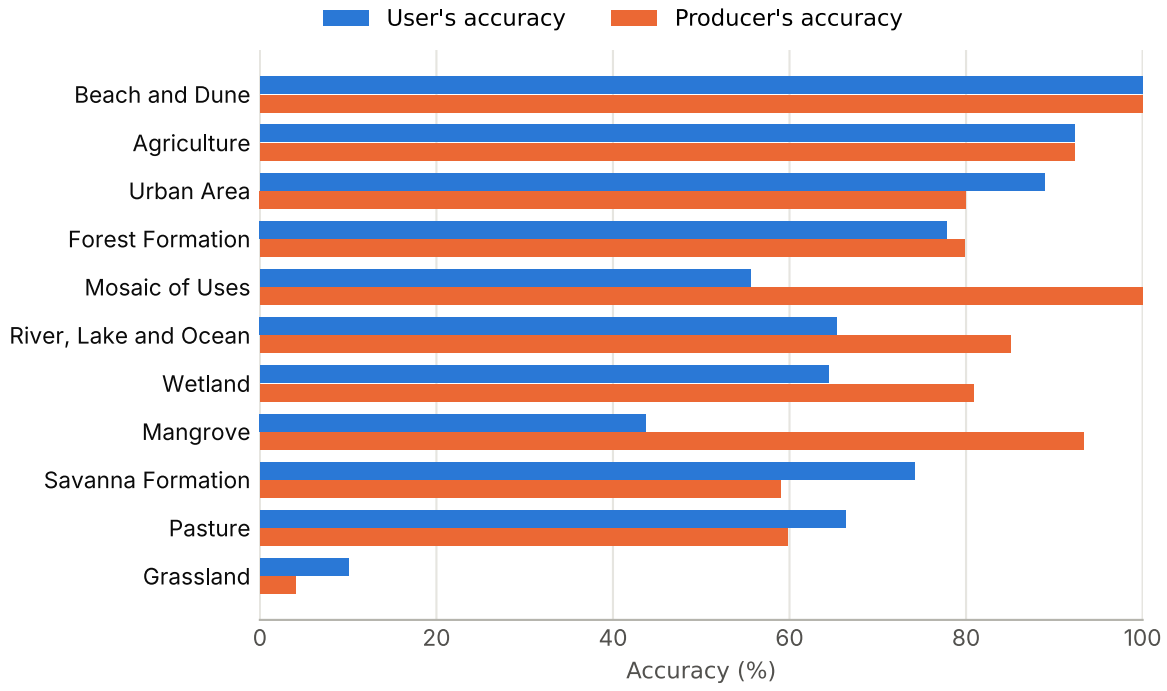
Forest Formation dominates the samples (282 of 655 reference samples) and is classified well, with user's and producer's accuracies near 80% (Table 1). Among the coastal classes that MapBiomas does not report, Beach and Dune was classified without errors, and Mangrove was rarely missed (one omission out of 15 reference samples) but often over-mapped: 18 of the 32 samples mapped as Mangrove were Forest Formation in the reference. Mosaic of Uses shows the same pattern, with every reference sample found but 8 of 18 mapped samples belonging to other classes. Grassland was the weakest class: only 1 of its 25 reference samples was mapped correctly.

Table 1. Samples and accuracy per level-2 class (sample counts).

			Samples		User's		Producer's	
ID	Class	MapBiomas name	Map	Reference	Accuracy	Commission	Accuracy	Omission
3	Forest Formation	Formação Florestal	289	282	78%	22%	80%	20%
4	Savanna Formation	Formação Savânica	93	117	74%	26%	59%	41%
5	Mangrove	Mangue	32	15	44%	56%	93%	7%
11	Wetland	Campo Alagado e Área Pantanosa	59	47	64%	36%	81%	19%
12	Grassland	Formação Campestre	10	25	10%	90%	4%	96%
15	Pasture	Pastagem	92	102	66%	34%	60%	40%
18	Agriculture	Agricultura	13	13	92%	8%	92%	8%
21	Mosaic of Uses	Mosaico de Usos	18	10	56%	44%	100%	0%
23	Beach and Dune	Praia e Duna	14	14	100%	0%	100%	0%
24	Urban Area	Área Urbanizada	9	10	89%	11%	80%	20%
33	River, Lake and Ocean	Rio, Lago e Oceano	26	20	65%	35%	85%	15%
	Total		**655**	**655**				

Map: samples assigned to the class by MapBiomas. Reference: samples labelled with the class by the interpreters.

Figure 1. User's and producer's accuracy per class, ordered by their mean.



3.2 Confusion matrix

Most errors involve the vegetation classes that dominate the region (Table 2). Forest Formation is confused mainly with Savanna Formation and Pasture, and it accounts for all of the over-mapping of Mangrove. Grassland reference samples were mapped mostly as Wetland (7), Pasture (6) and Savanna Formation (4).

Table 2. Confusion matrix (number of samples). Rows: MapBiomas map; columns: reference. Shaded cells are correctly classified samples.

ID	Map class	3	4	5	11	12	15	18	21	23	24	33	Total
3	Forest Formation	**225**	34	1	0	3	23	1	0	0	0	2	**289**
4	Savanna Formation	12	**69**	0	0	4	8	0	0	0	0	0	**93**
5	Mangrove	18	0	**14**	0	0	0	0	0	0	0	0	**32**
11	Wetland	8	0	0	**38**	7	6	0	0	0	0	0	**59**
12	Grassland	0	6	0	0	**1**	2	0	0	0	0	1	**10**
15	Pasture	16	4	0	3	6	**61**	0	0	0	2	0	**92**
18	Agriculture	0	0	0	0	0	1	**12**	0	0	0	0	**13**
21	Mosaic of Uses	1	4	0	0	3	0	0	**10**	0	0	0	**18**
23	Beach and Dune	0	0	0	0	0	0	0	0	**14**	0	0	**14**
24	Urban Area	0	0	0	0	0	1	0	0	0	**8**	0	**9**
33	River, Lake and Ocean	2	0	0	6	1	0	0	0	0	0	**17**	**26**
	Total	**282**	**117**	**15**	**47**	**25**	**102**	**13**	**10**	**14**	**10**	**20**	**655**

Class IDs as in Table 1.

3.3 Overall accuracy and comparison with Brazil

Overall accuracy in the study area was 71.6%, well below the 85.8% that MapBiomas reports for Brazil (Table 3). The larger share of the gap comes from allocation disagreement, 20.8% against 9.4% nationally: the maps get the proportions of the classes roughly right but place them in the wrong locations more often than in the country as a whole. The Kappa of 0.62 corresponds to substantial agreement (Landis and Koch, 1977). The comparison is only indicative, because the national figures are area-weighted estimates (Stehman, 2014) over a set of classes that differs from the one assessed here.

Table 3. Overall metrics for the study area and for Brazil.

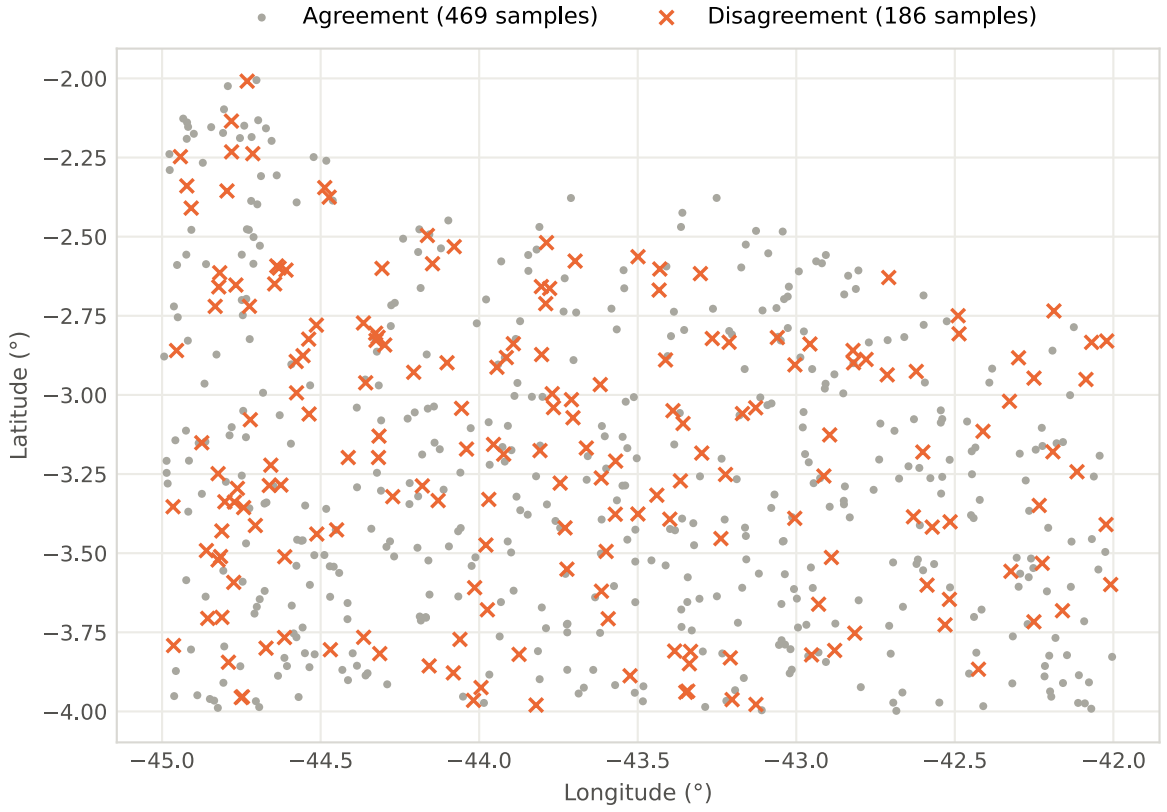
Metric	Study area (SA-23-Z)	Brazil (MapBiomas)
Overall accuracy	71.6%	85.8%
Overall error	28.4%	14.2%
Quantity disagreement	7.6%	4.8%
Allocation disagreement	20.8%	9.4%
Cohen's Kappa	0.62	—

Study area: sample counts, 655 samples. Brazil: MapBiomas accuracy statistics for Collection 8 (MapBiomas, 2023), area-weighted; MapBiomas does not report Kappa.

3.4 Spatial distribution of disagreement

The 186 samples where the map and the reference disagree are spread over the whole chart group, without a visible concentration in any sub-region (Figure 2). This suggests that the errors are not driven by one problematic area, such as the coast or a single scene.

Figure 2. Validation samples in chart group SA-23-Z (WGS 84), by agreement between the MapBiomas map and the reference.



4. Discussion and limitations

Coastal classes. Mangrove is reliable where the reference says Mangrove, but the map extends it into neighbouring forest, which inflates its area. Separating Mangrove from Forest Formation is a known difficulty and deserves more samples and review by interpreters who know the region. The perfect score for Beach and Dune rests on only 14 samples, all in the Lençóis Maranhenses, where the dunes are hard to confuse with other covers. Apicum, a class of interest in the region, could not be assessed at all.

Grassland. With a producer's accuracy of 4%, Grassland is the least accurate class in the study area, below the about 9% that MapBiomas reports for Brazil.

Sample counts versus area. All results above are computed from sample counts. An area-weighted estimate, computed in the analysis notebook from the inclusion probabilities of the samples, gives an overall accuracy of 75.6%. The national MapBiomas statistics are area-weighted, which is one more reason to read Table 3 as indicative.

Training samples. 82 of the 712 samples (69 of the 655 analysed) are flagged in the MapBiomas data as having been used to train the classifier. The MapBiomas assessment code is meant to discard them, but its filter does not match the label used in the data, so they remain in this analysis, as in the published article. Excluding them changes overall accuracy from 71.6% to 72.0% and does not change Kappa at two decimal places, so the conclusions hold.

Representativeness. The chart group covers about 72% of the region, and some classes have few samples. Per-class figures for classes with fewer than about 20 samples should be read with caution.

5. Conclusions

At legend level 2, MapBiomas Collection 8 reaches an overall accuracy of 71.6% in the São Luís intermediate region, with a Kappa of 0.62. This is substantial agreement, but clearly below the national figures, mostly because of allocation errors. The assessment adds accuracy figures for classes that MapBiomas does not report: Beach and Dune is mapped very reliably, while Mangrove is over-mapped at the expense of Forest Formation. A dedicated set of reference samples for the coast, validated by local experts, is needed to assess Mangrove, Apicum and Beach and Dune with more confidence.

Reproducibility

Code, data and both notebooks are in github.com/LambdaGeo/acuracia-slz (MIT license for the code; the data derive from MapBiomas and follow its terms of use). With Python 3.12 and the pinned dependencies in `requirements.txt`, running `2_acuracia_sa23z.ipynb` reproduces the published results, and `python build_report.py` rebuilds this report. The report checks the values quoted in its text against the analysis and stops if they differ.

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